

Episodic memory rescues working memory via pattern separation, pattern completion, and predictive recall

Zihan Bai

First-year talk, 9/25/26

Background



Project goals

Prior literature suggests EM plays a role in WM tasks

Beukers et al, *TiCS*, 2021; Hoskin et al., *CABN*, 2019; Bartsch & Oberauer, *Cognition*, 2023

How might EM support WM?

Pattern Separation



Yassa & Stark,
Trends Neurosci., 2011

Predictive recall



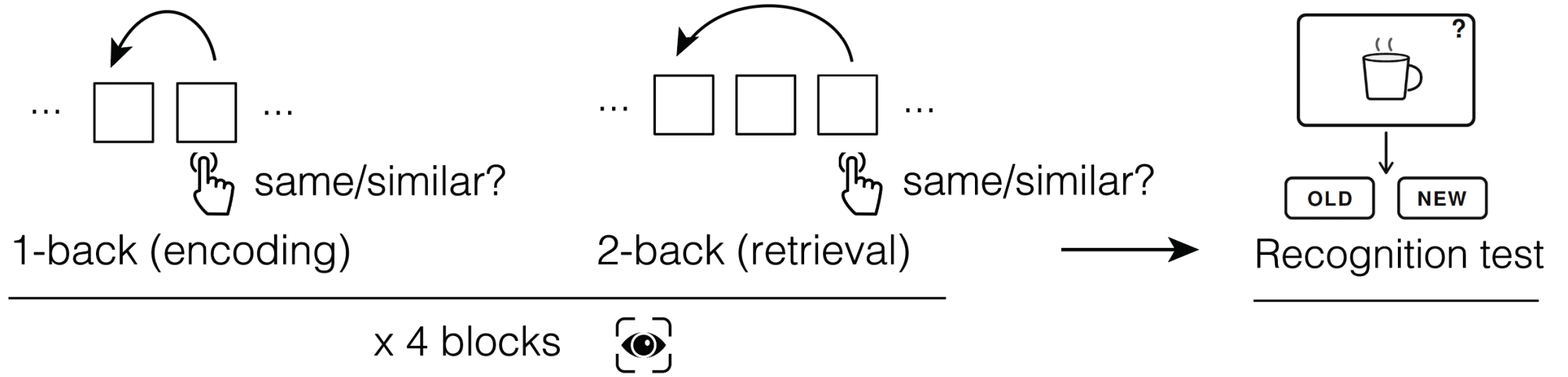
Michelmann et al.,
Nat. Comm., 2021

Pattern Completion



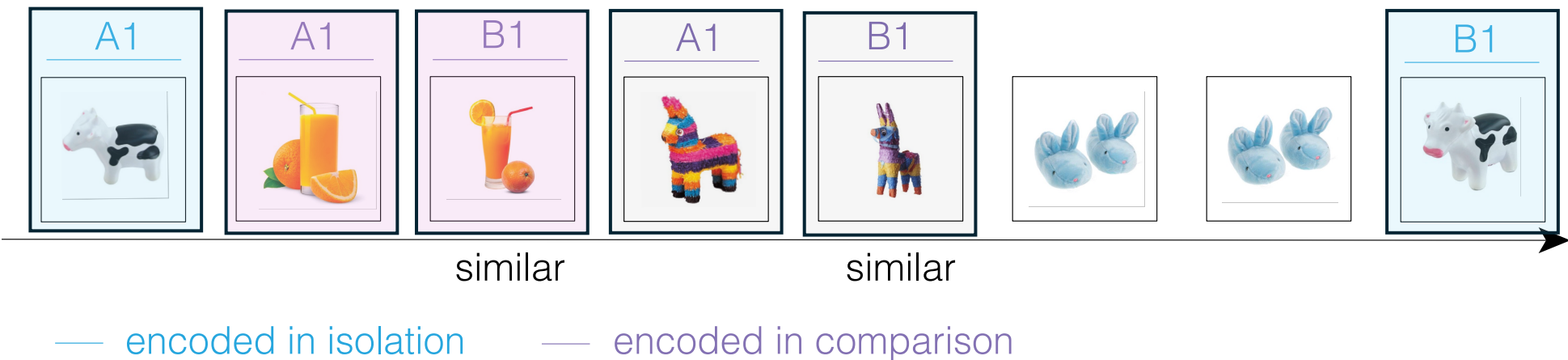
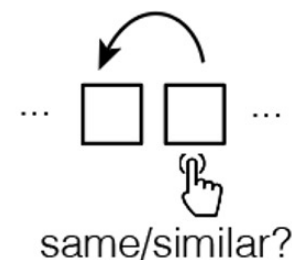
Wynn et al.,
PNAS, 2020

Experiment: the MST-Back task

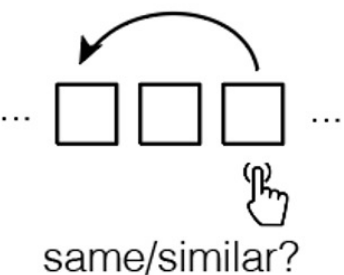


Experiment: the MST-Back task

1-back (encoding) task



2-back (retrieval) task



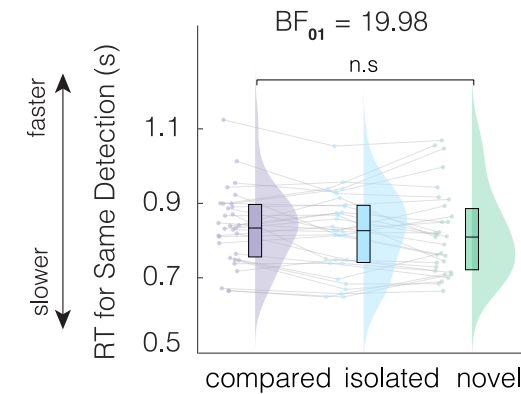
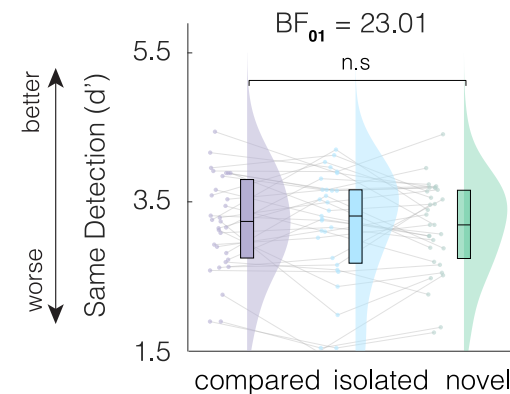
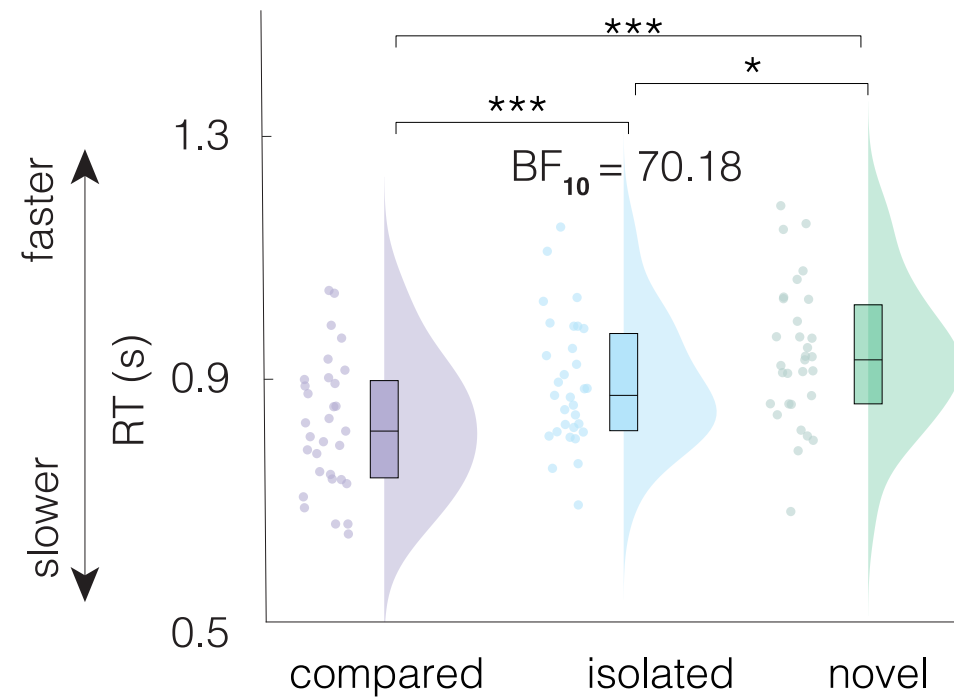
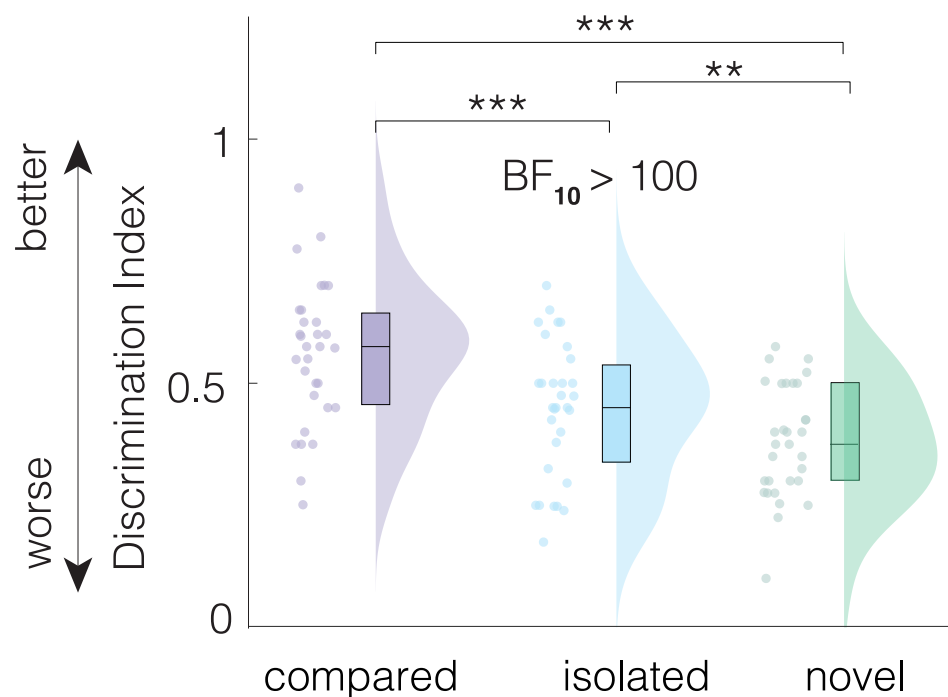
A key feature of our task

The novelty of an image **is not predictive** of the time or type of response.

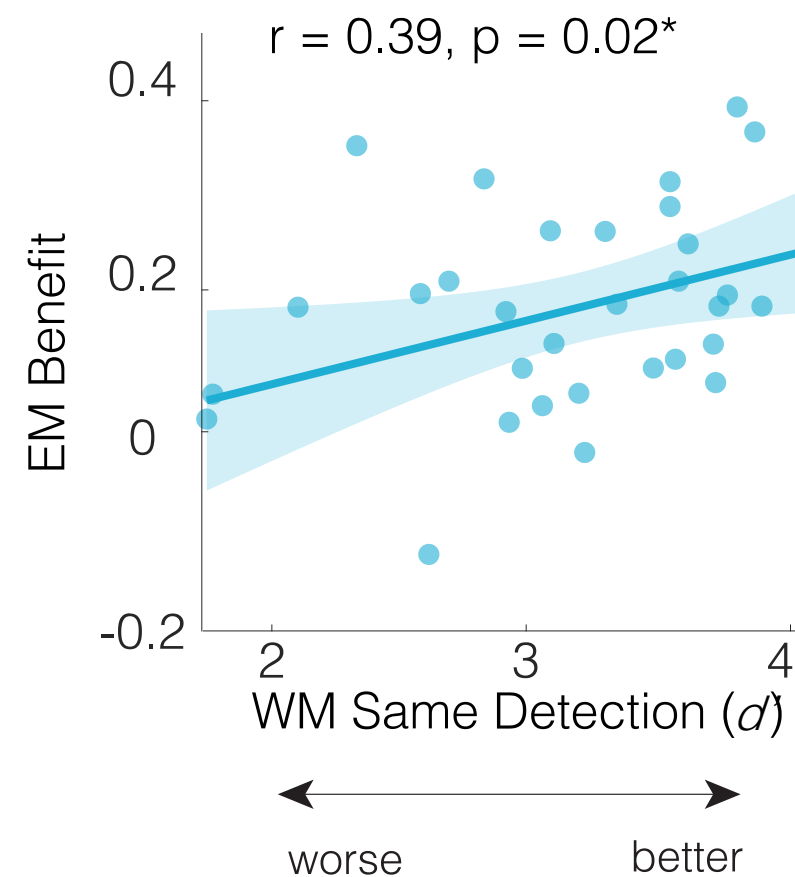
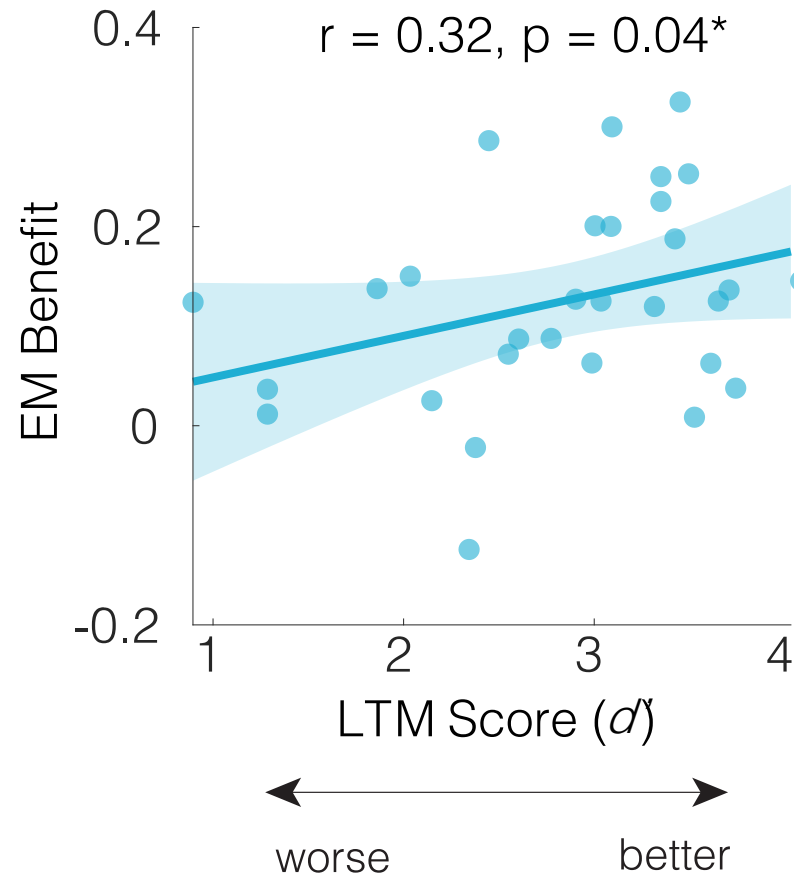
Consequently, EM can neither reduce WM load, nor can episodic representations replace WM representations.

Instead, object representations stored in EM **need to be integrated** with the content of WM in order to improve performance.

A graded, item-specific episodic benefit for WM



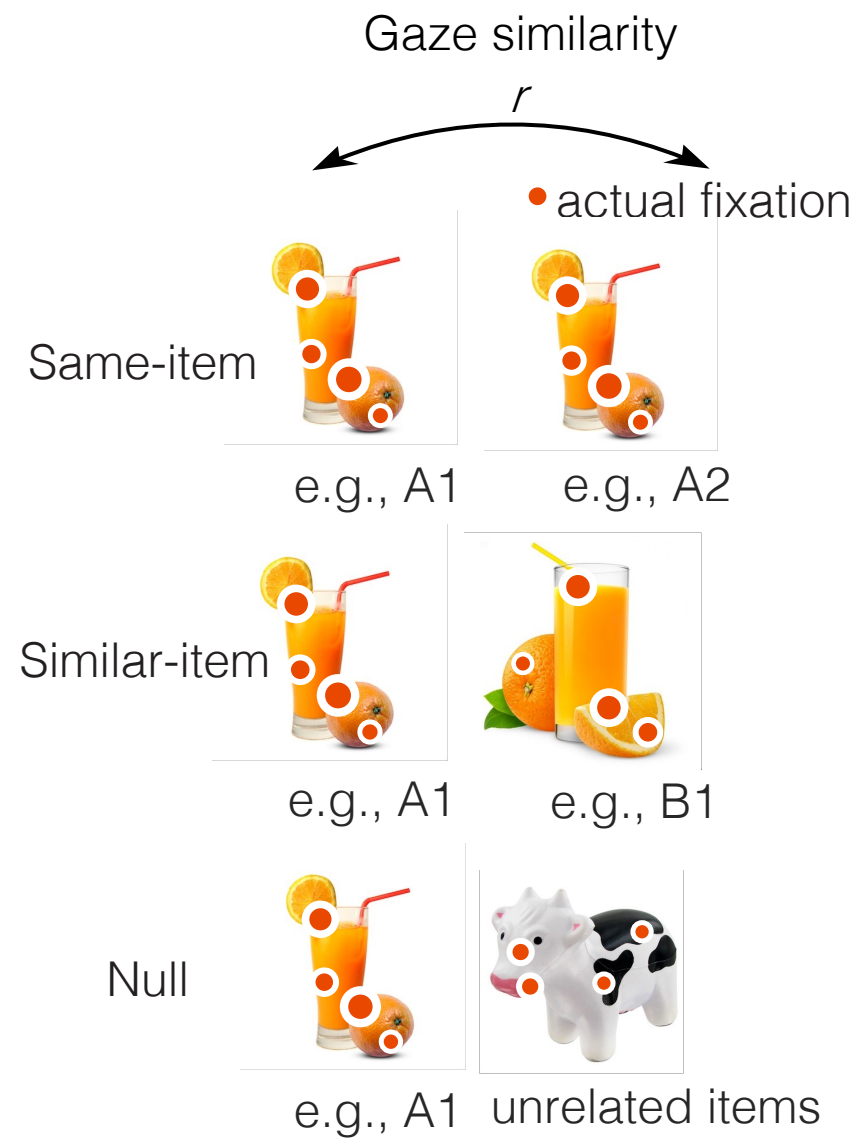
The benefit reflects a shared memory factor



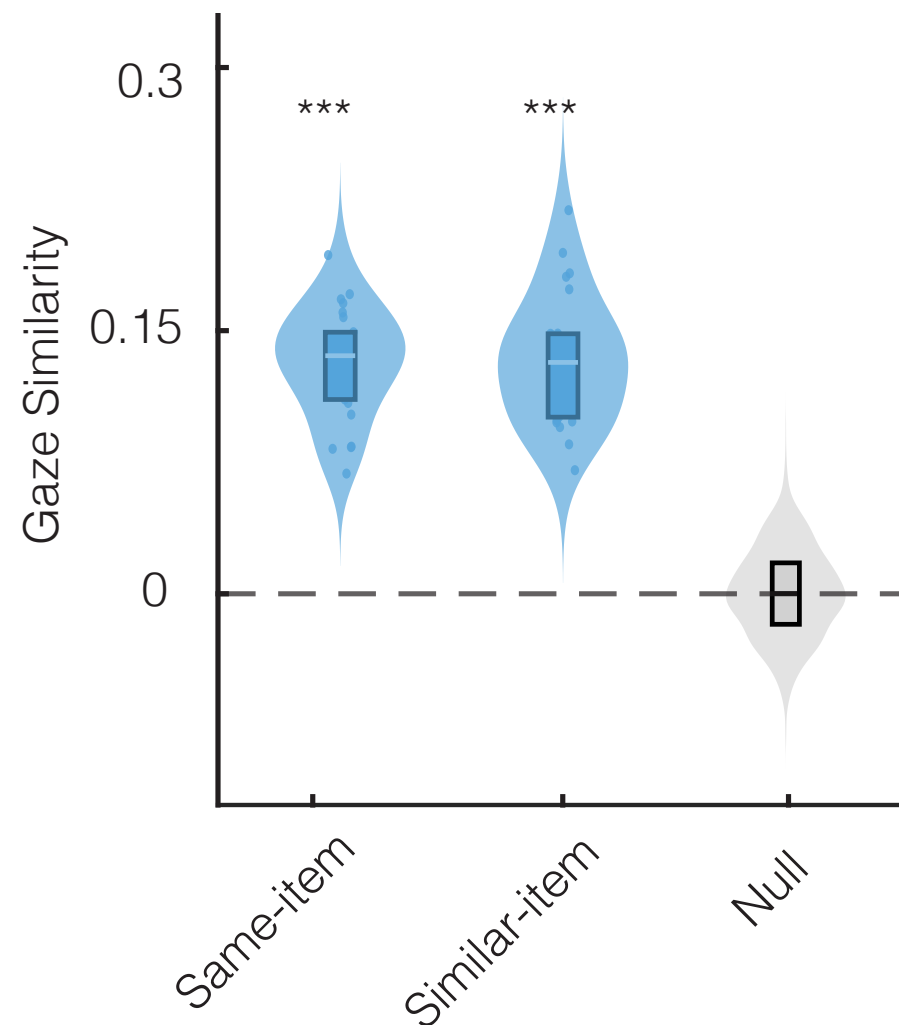
Greater EM benefit for subjects with better LTM

Greater EM benefit for subjects with better WM

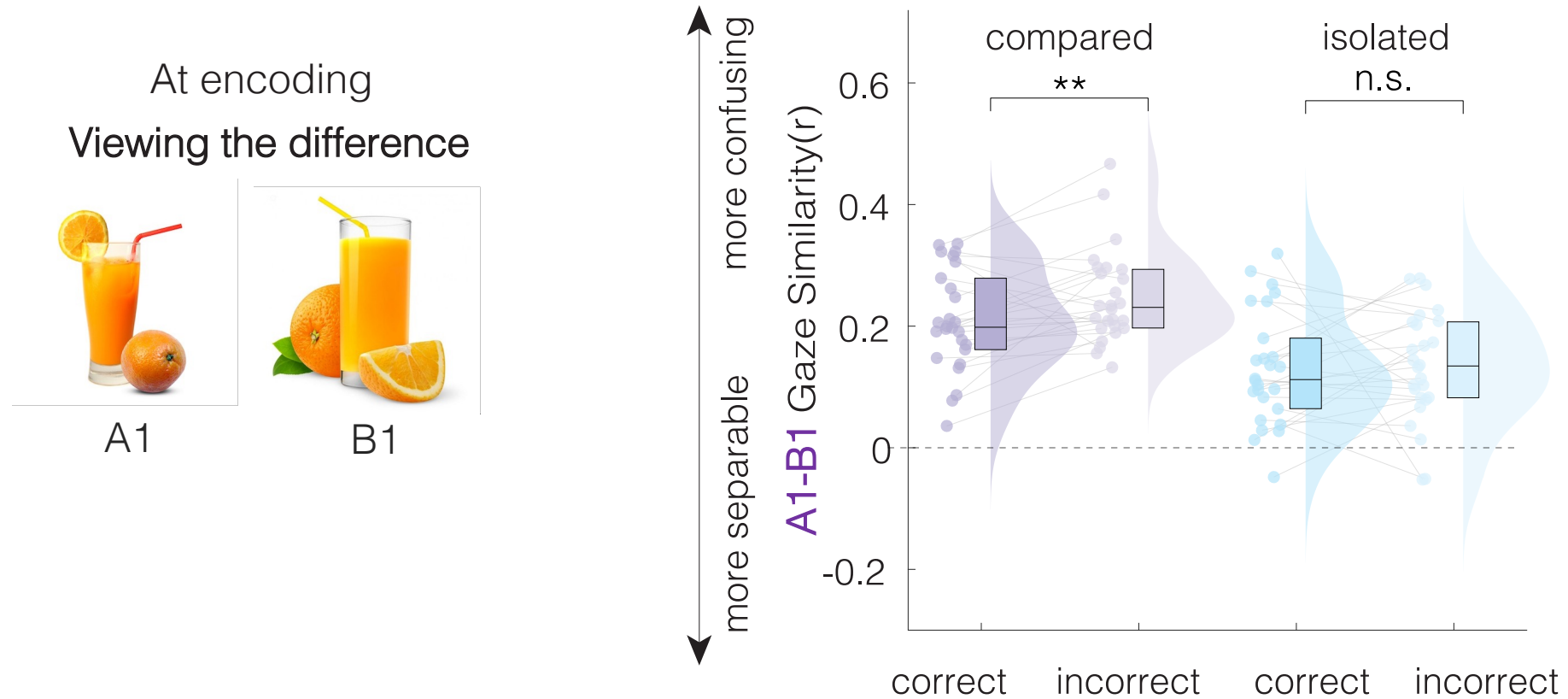
Test the mechanisms of EM that supported WM



Gaze similarity tracks image similarity

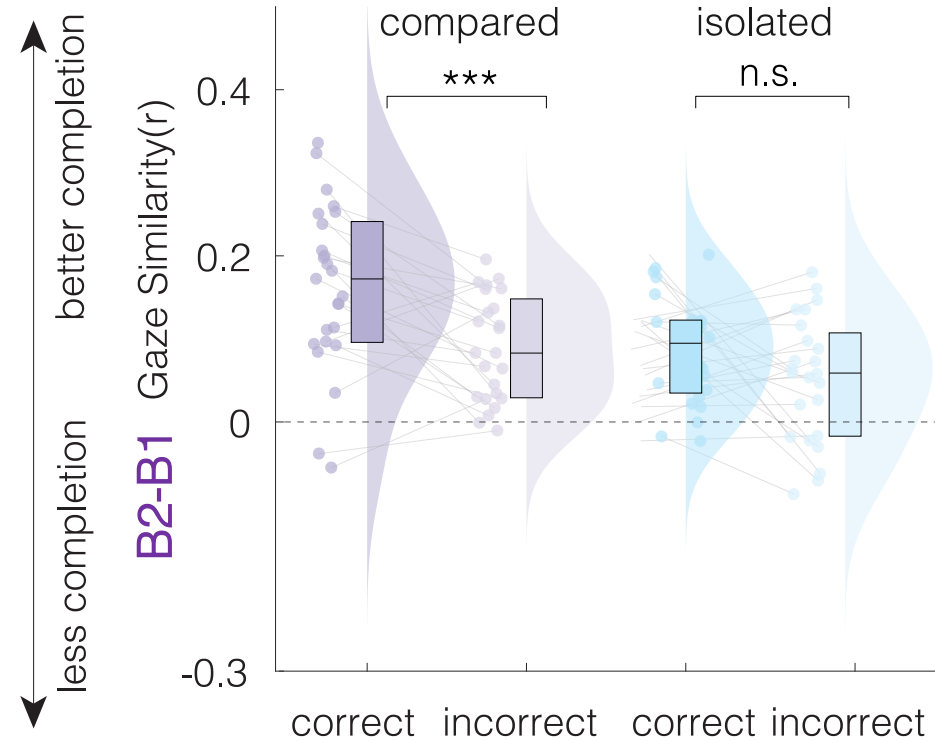
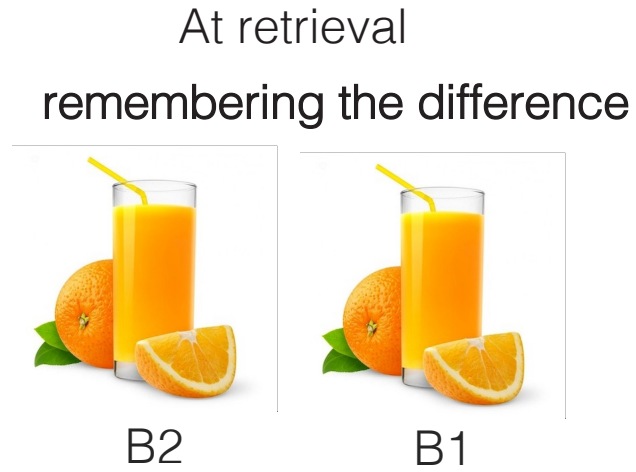


Gaze-pattern separation predicts subsequent discrimination success



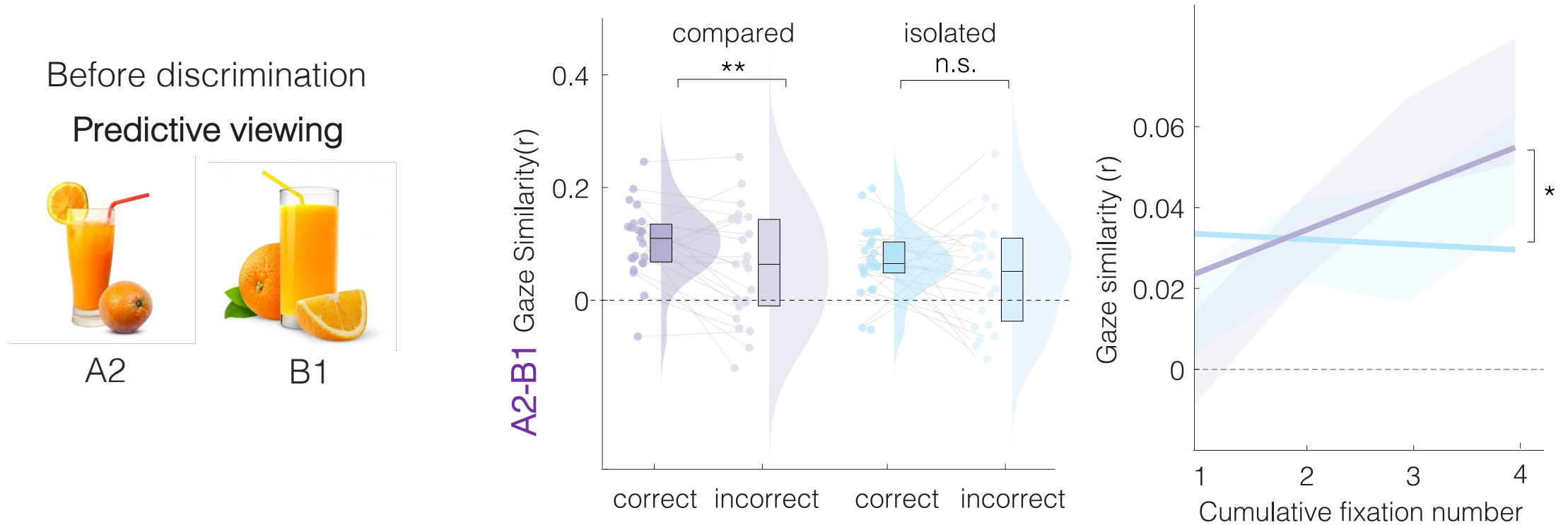
Pairs encoded with more discriminative gaze (less similar) in 1-back were better subsequently discriminated in 2-back

Gaze-pattern completion at retrieval marks discrimination success



Diagnostic gaze patterns were recapitulated during successful discrimination

Predictive recall of gaze patterns predict upcoming discrimination success

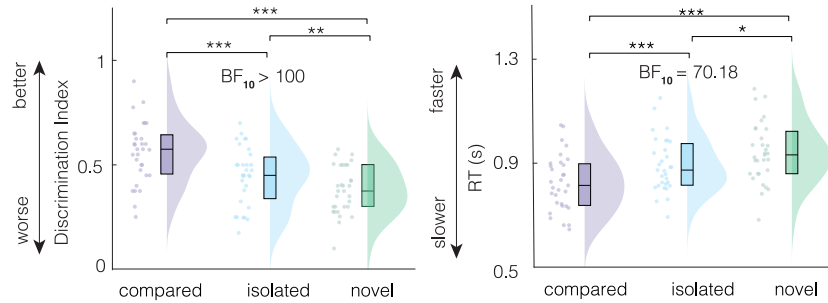


Two trials before the target appeared, predictive viewing of the first of two pairmate foreshadowed upcoming discrimination success (predictive recall of diagnostic features)

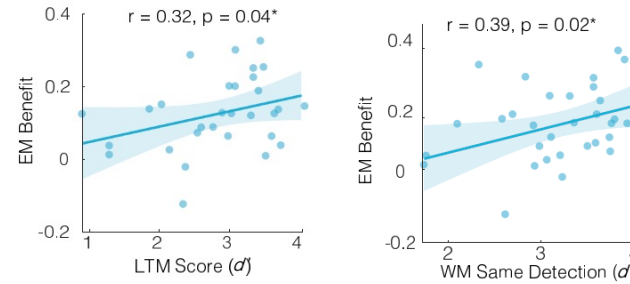
Predictive recall gradually emerged with more fixations

Conclusion

Evidence for an episodic benefit for WM

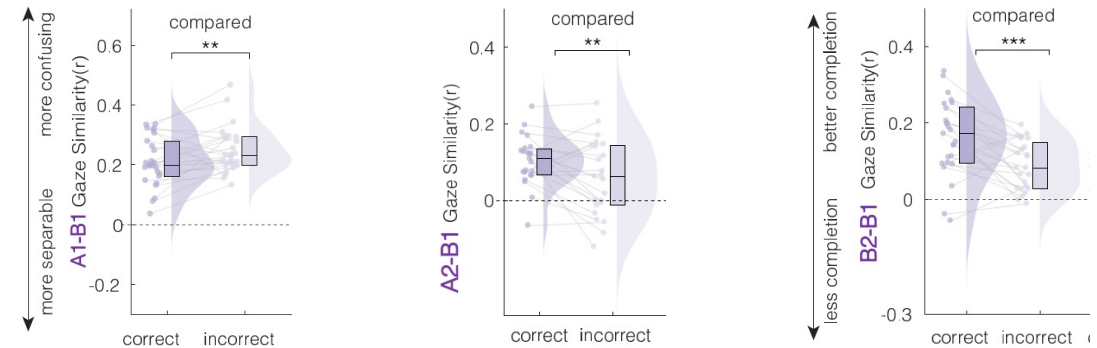


A shared factor contributing to EM-WM interaction



Three hallmark computations underlies *when* and *how* this performance benefit emerges

- (i) Diagnostic memory traces are laid down at encoding
- (ii) Upcoming content is predicted before discrimination
- (iii) And it is reinstated during discrimination at retrieval



Our findings characterize the mechanism of integration between episodic and working memory

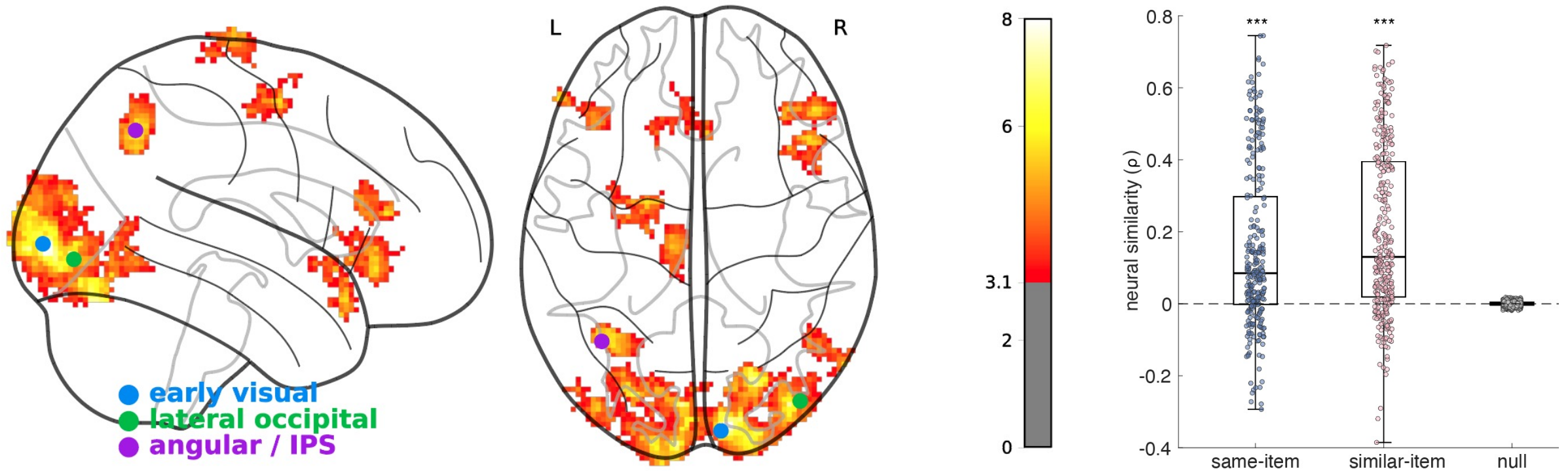
Future direction

To characterize the neural mechanisms of interaction between episodic and working memory

Specifically, **where** in the brain, and at **which moments** EM that supported WM discrimination emerged

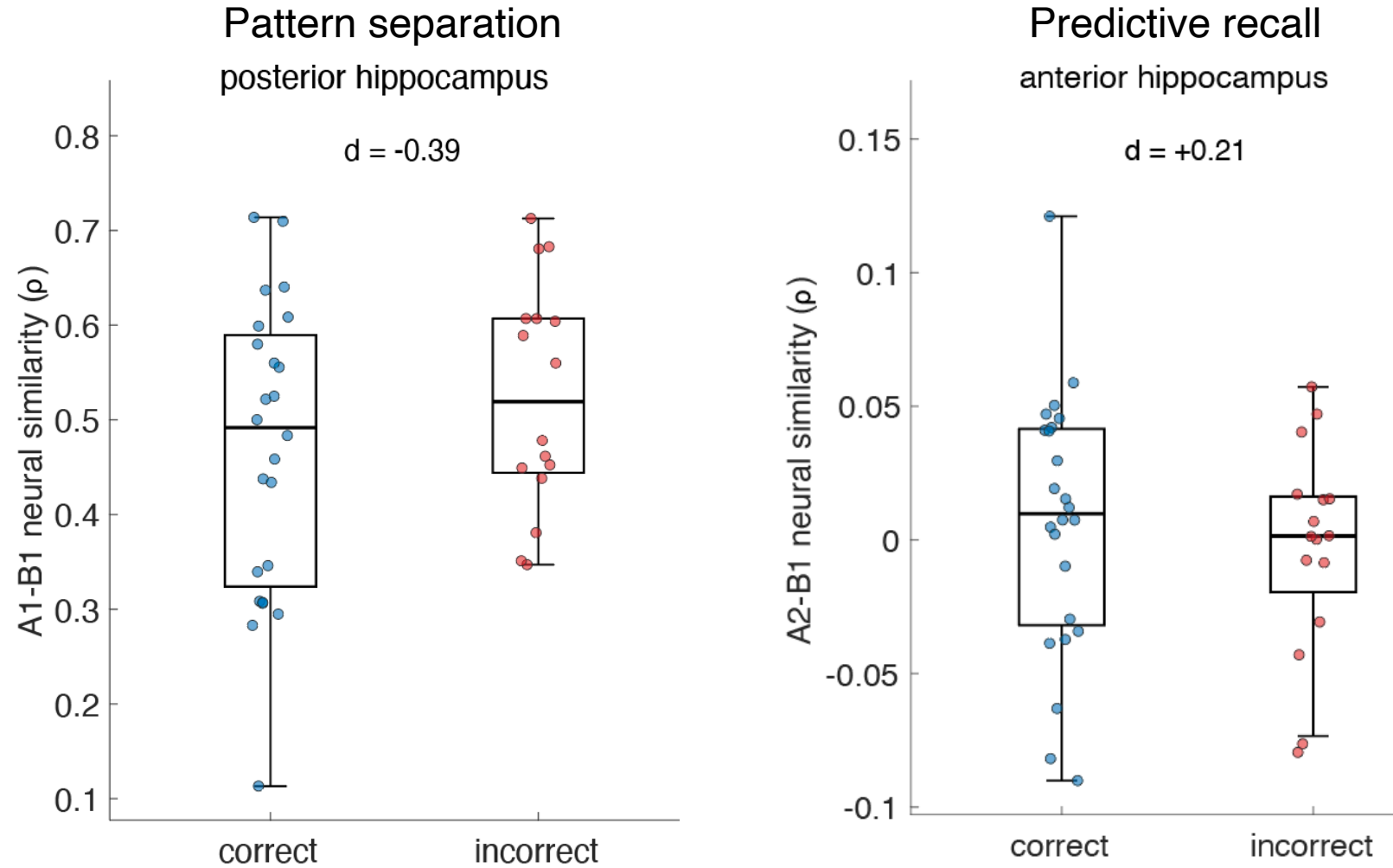
Ongoing high-res fMRI study (N=1)

Neural similarity tracks image similarity (cluster-corrected ($z > 3.09$, $p < .05$))



Future direction

Hippocampal pattern separation and predictive recall tracks discrimination success





The Interactive Dynamics of Episodic Memory Lab



Fougnie Lab

New York University Abu Dhabi



Sebastian Michelmann



Daryl Fougnie



Check out our new preprint!



zihan.bai@nyu.edu



zihanbai.bsky.social